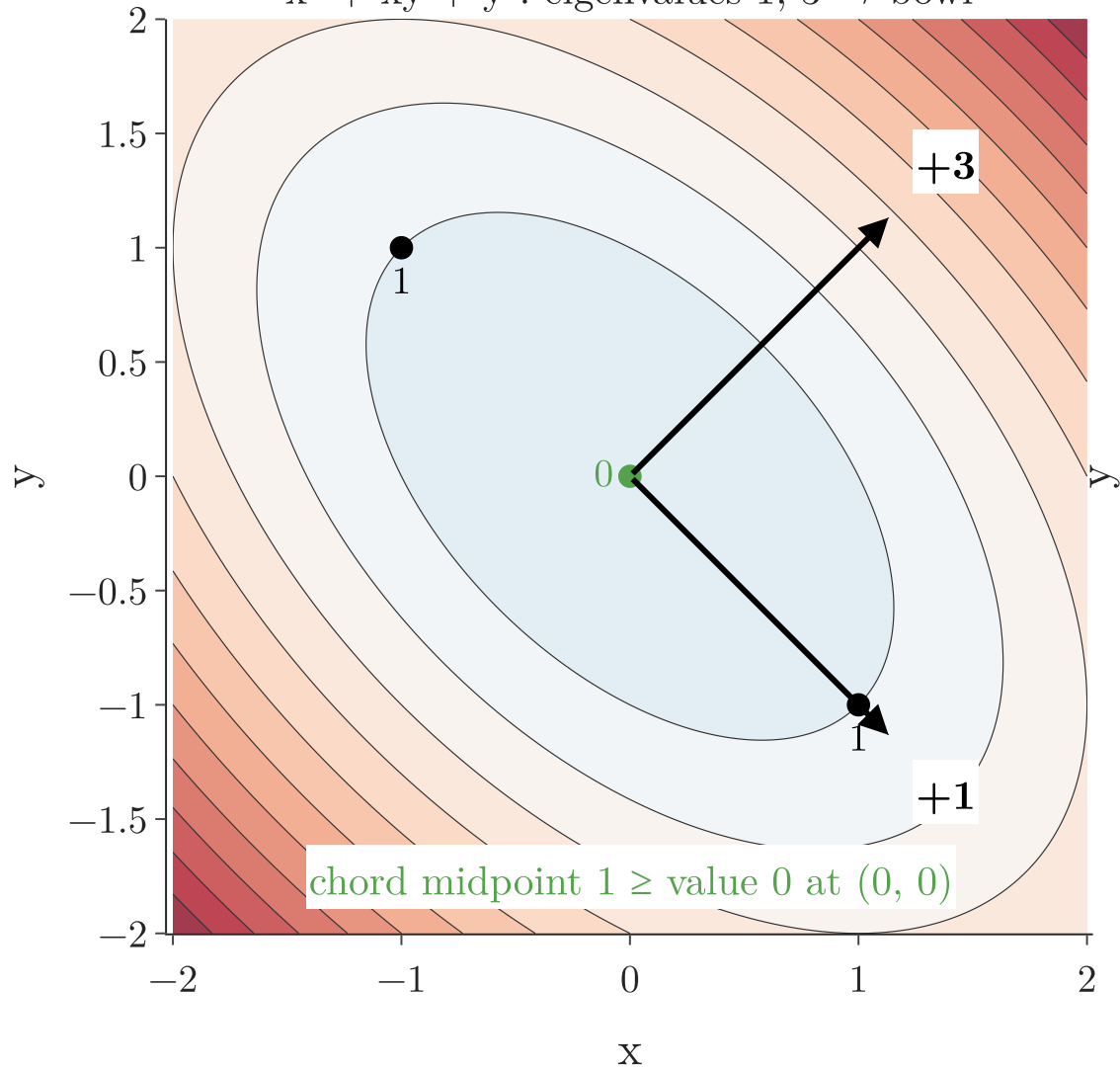


$x^2 + xy + y^2$: eigenvalues 1, 3 $\rightarrow$ bowl



$x^2 + 3xy + y^2$: eigenvalues 5, -1 $\rightarrow$ saddle

